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ABSTRACT

Measuring the internal quality factor of coplanar waveguide superconducting resonators is an established method of determining small losses in superconducting devices. Often, resonator losses are attributed to two-level system (TLS) defects using a power dependent model for the quality factor. However, excess non-equilibrium quasiparticles can also limit the quality factor of superconducting resonators. Here, a two-temperature, power and temperature dependent model is used to evaluate resonator losses for titanium nitride and aluminum coplanar waveguide resonators with loss tangents of 10^{-5} – 10^{-6} . This model enables a set of parameters to describe both the internal quality factor and fractional frequency shift of the resonator across a broad range of average cavity photon numbers, from 1 to 10^6 , and temperatures between those limited by the experimental apparatus near 50 mK and thermal quasiparticle dominated. The presented model also offers potential insight into observing the temperature and power dependence of TLS–TLS interactions in these resonators.

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I. INTRODUCTION

Low-loss superconducting circuit elements are critical for high coherence superconducting quantum circuits. Coplanar waveguide (CPW) resonators are an essential component for qubit devices and are used as readout elements in two-dimensional transmon^{1–3} and flux^{4,5} qubits. CPW resonators also provide value for material characterization.^{6–9}

Superconducting quantum circuits suffer from microwave loss originating from two-level system (TLS) defects,^{10,11} nonequilibrium quasiparticles,^{12–14} vortices,^{15,16} microwave radiation,¹⁷ and coupling with package modes.¹⁸ These loss mechanisms have a characteristic dependence on applied microwave power, bath temperature, magnetic field, frequency of operation, and resonator geometry.

Conventionally, the power dependence of the internal quality factor, Q_i , of CPW resonators is measured at the lowest possible bath temperature. This measurement approach is mostly suited for devices that are limited by TLS loss, and the observation of high power and low power saturation enables quantification of the TLS

loss. The TLS loss can be difficult to assess if loss saturation at high powers is not observed.¹⁹ Unfortunately, other loss mechanisms are difficult to isolate from power-dependent measurements at a single temperature. Of particular concern is the loss contribution from non-equilibrium quasiparticles that is difficult to quantify²⁰ but may be comparable to TLS losses for some refined resonator materials and fabrication processes.

While TLS loss is relatively straightforward to calculate, modeling non-equilibrium quasiparticle loss is calculated by solving coupled nonlinear energy-dependent rate equations of phonon and quasiparticle systems.^{20–23} For the fully self-consistent nonlinear model, the energy-dependent equations require several poorly known or unobservable material-dependent parameters. In addition, these models are most often applied in high-power regimes where quasiparticle losses are expected to dominate instead of the high-quality material and low-power regime that is relevant to quantum information devices.

This article describes a two-temperature and power dependent model that simultaneously considers TLS and non-equilibrium

quasiparticle losses in a superconducting resonator. The TLS loss is calculated using the model expressed in terms of the lifetime and decoherence time of an interacting TLS ensemble.²⁴ The quasiparticle loss is calculated by representing the quasiparticle number density with an effective temperature that may differ from the bath temperature. This approximation is suitable to describe thermalized, non-equilibrium quasiparticle behavior.²⁵ In addition, a single electron relaxation rate is included in the quasiparticle rate equations to provide an energy pathway to restore equilibrium between the effective quasiparticle and bath temperatures without recombination. Temperature and power dependent loss measurements of high-quality factor aluminum and titanium nitride thin film resonators are used to illustrate the effectiveness of this model to identify the TLS and quasiparticle contributions to the measured microwave loss.

II. RESONATOR MODEL

A. Two-level systems

At low temperatures and low powers, unsaturated TLS defects resonantly interact with the microwave electric field in the waveguide resonator. This interaction leads to energy transfer to the phonon population, resulting in losses. For the resonators studied here that operate in the 5 GHz range, the resonant energy is on the order of 20 μeV . This energy is small compared to the accuracy of current numerical methods that might be used for the simulation and identification of the precise microscopic structure of TLS defects.²⁶ However, it is known to be associated with dielectrics adjacent to or near superconducting materials such as oxides,¹¹ hydroxyl groups,²⁷ or chemical residues.²⁸ In coplanar waveguide resonators grown on low-loss substrates, TLS defects primarily reside at the metal–air, metal–substrate, and substrate–air interfaces.²⁹ Not only do TLS defects interact with the resonant field, but they also interact with each other, creating a complex dynamic that may be heavily influenced by the specific details of a given sample.¹¹

The typical observation of the measured quality factor being directly proportional to power is an observation of the degree of saturation of an interacting ensemble of TLS defects. The loss is maximum at low power and low temperature when the TLSs are primarily in the ground state and decreases with either increasing temperature or power when the upper state TLS occupancy increases, thus reducing the ability of the TLS ensemble to resonantly absorb microwave photons. The temperature dependence using tanh and the power dependence using a power-law are well supported throughout the literature. The temperature and power dependent loss, Q_{TLS} , of an ensemble of TLSs, including the TLS–TLS interaction, can be expressed as²⁴

$$\frac{1}{Q_{\text{TLS}}(\omega_r, n, T)} = \frac{1}{Q_{\text{TLS}}^0} \left(\frac{\tanh \frac{\hbar\omega_r}{2k_B T}}{\sqrt{1 + \left(\frac{n\beta_2}{DT\beta_1} \right) \tanh \frac{\hbar\omega_r}{2k_B T}}} \right). \quad (1)$$

The contribution of the TLS defects to the dielectric constant is reflected by the change in the resonance frequency. The temperature dependent fractional resonance frequency shift is given by^{10,11,24}

$$\frac{\delta f(\omega_r, T)}{f_r} = \frac{1}{\pi Q_{\text{TLS}}^0} \left[\Re \left(\Psi \left(\frac{1}{2} + \frac{1}{2\pi i} \frac{\hbar\omega_r}{k_B T} \right) \right) - \log \frac{\hbar\omega_r}{2\pi k_B T} \right]. \quad (2)$$

In these equations, the TLS loss coefficient, $(Q_{\text{TLS}}^0)^{-1} = \sum_i p_i \tan \delta_i$, is determined from the sum of the loss tangent, $\tan \delta_i$, and participation ratio, p_i , of the different regions. In Eq. (2), $\Re(\Psi(x))$ is the real part of the complex digamma function evaluated at x . For coplanar waveguide resonators, the individual contributions from different dielectric materials, such as the native oxide at the metal–air interface and substrate–air interface, dielectrics at the metal–substrate interface, or in the substrate, are considered. Conventional terms of TLS behavior are the geometric resonance frequency of the resonator, $f_r = \omega_r/(2\pi)$; temperature, T ; the average number of microwave photons in the cavity, n ; and β_2 , which defines the strength of the saturation. In lieu of the single critical photon number, we follow the expression presented by Crowley *et al.*²⁴ $n_c \propto 1/T_{1\text{TLS}} T_{2\text{TLS}}$, where the average lifetime of an interacting TLS ensemble is expressed as $T_{1\text{TLS}} \propto \tanh(\hbar\omega_r/2k_B T)$, and the decoherence time is $T_{2\text{TLS}} \propto (DT\beta_1)^{-1}$. The variables associated with describing the decoherence of the TLS ensemble, D and β_1 , are empirical parameters.

The complex temperature and power dependence of TLS defects in resonators has primarily been explored through measurements of the quality factor. Many studies experimentally observe significant variations between resonators on a given chip. The theoretical underpinnings of loss may be described by several models of ensembles of interacting TLS defects considering tunneling³⁰ and dipole interactions¹¹ as well as geometric contributions that can only be incorporated through extensive electric field simulations.³¹ The power dependence captured through β_2 is widely captured as a constant that is adjusted for a given resonator. However, the two decoherence parameters, D and β_1 , are not well studied. It is reasonable to concede that the chosen description for TLS decoherence is insufficiently described by the simple power model. Therefore, D and β_1 are independently determined for each power.

B. Quasiparticles

In superconductors, quasiparticles provide a resistive channel. According to the BCS theory,³² the thermal quasiparticle density vanishes at low temperatures compared to the critical temperature of the superconductor. This is an important motivation for operating superconducting qubits at low-millikelvin temperatures despite the common usage of materials that have a critical temperature, T_c , around 1 K. However, there are experimental observations of limited performance of superconducting quantum devices that are attributed to excess quasiparticles despite sample measurement temperatures near 10 mK (1% of T_c of aluminum).^{33–36}

Theoretically, any photon or phonon with energy greater than the superconducting gap (2Δ) can break Cooper pairs and generate quasiparticles. These pair-breaking particles may originate from stray infrared, microwave, or optical photons;¹² cosmic or gamma rays;¹³ spurious antenna modes;¹⁴ or non-thermalized input signals having excess high-energy photons.³⁷ Even sequential multiphoton absorption of sub-gap energy microwave signals can lead to quasiparticle excitation. Their single-particle relaxation from an excited energy state to the band edge can generate sufficiently high energy phonons to break Cooper pairs.^{20,21} These effects are captured in the term representing external excitation, I_{ext} .

Calculating the energy averaged number density of net quasiparticles, n_{qp} , can be represented as a two-temperature rate equation

$$\frac{\partial n_{qp}(T_{qp})}{\partial t} = I_{ext} + \Gamma_G N_{2\Delta}(T_b) - \Gamma_R n_{qp}^2(T_{qp}) - \Gamma_S (n_{qp}(T_{qp}) - n_{qp}(T_b)), \quad (3)$$

with terms that represent quasiparticle generation through Cooper pair breaking by high energy phonons, $N_{2\Delta}$, quasiparticle relaxation into Cooper pairs, and relaxation of excited quasiparticles toward thermal equilibrium. Here, Γ_G is the pair breaking rate, Γ_R is the quasiparticle recombination rate, and Γ_S is the single quasiparticle relaxation rate that returns the quasiparticle to thermal equilibrium. The two temperatures, T_b and T_{qp} , represent the bath temperature and effective quasiparticle temperature.

The key assumption of the effective temperature model in Eq. (3) is that the population density of each particle-type is adequately described by the appropriate thermal distribution such that the complete microscopic model^{21,38} does not need to be used. This assumption is valid when the intraparticle scattering rate is larger than the interparticle scattering rate. The effective temperature model allows time-varying non-equilibrium conditions to be described, but it is not suitable for strong continuous excitations that cause spectral hole burning or other athermal population distributions, or pulsed dynamics that are on a timescale of the intraparticle scattering rates.

Under conditions that the superconductor achieves thermal equilibrium, there is no external energy coupled into the quasiparticle population, and the quasiparticles are in thermal equilibrium with phonons, i.e., $I_{ext} = 0$ and $T_{qp} = T_b$. The quasiparticle and the 2Δ phonon number densities simplify to

$$\Gamma_G N_{2\Delta}(T_b) = \Gamma_R n_{qp}^2(T_b). \quad (4)$$

However, when equilibrium conditions are not met, such as when quasiparticles are excited by a modest external energy source ($I_{ext} \neq 0$), the quasiparticles may no longer be in thermal equilibrium with the bath and are more accurately described by an effective quasiparticle temperature that is above the bath temperature ($T_{qp} > T_b$). The external energy may be blackbody energy coupled into the superconducting resonator through the rf lines or a leaky sample enclosure. In the presence of a constant (continuous wave) external energy source, the steady-state relationship between the quasiparticle and the pair-breaking phonon number densities becomes

$$\Gamma_R (n_{qp}^2(T_{qp}) - n_{qp}^2(T_b)) + \Gamma_S (n_{qp}(T_{qp}) - n_{qp}(T_b)) = I_{ext}. \quad (5)$$

The positive root is the quasiparticle number density with an elevated quasiparticle temperature ($T_{qp} > T_b$),

$$n_{qp}(T_{qp}) = \sqrt{\frac{I_{ext}}{\Gamma_R} + \left(n_{qp}(T_b) + \frac{\Gamma_S}{2\Gamma_R}\right)^2} - \frac{\Gamma_S}{2\Gamma_R}. \quad (6)$$

The Fermi–Dirac distribution function in the low particle density limit, $f(E, T_{qp}) = \exp(-E/k_B T^*)$, where E is the quasiparticle

excitation energy, uses the effective temperature, T^* , to describe either the non-equilibrium quasiparticle temperature, T_{qp} , or equilibrium quasiparticle temperature, T_b . A thermalized quasiparticle number density³⁹ can be expressed as

$$n_{qp}(T^*) = 2N(0)\sqrt{2\pi k_B T^* \Delta(T^*)} \exp\left(-\frac{\Delta(T^*)}{k_B T^*}\right). \quad (7)$$

Here, $N(0)$ is the normal state single-spin density of states at the Fermi level and k_B is Boltzmann's constant. For a superconductor with a critical temperature, T_c , at low-temperatures ($T^*/T_c < 0.3$), the temperature dependent superconducting gap from BCS theory is approximated as⁴⁰

$$\Delta(T^*) \approx \Delta(0) \exp\left[-\sqrt{\frac{2\pi k_B T^*}{\Delta(0)}} \exp\left(-\frac{\Delta(0)}{k_B T^*}\right)\right], \quad (8)$$

where the superconducting gap at $T = 0$ K is $\Delta(0) \approx 1.76 k_B T_c$. When $T^* = T_b$, the quasiparticle number density represents thermally generated quasiparticles, and when $T^* = T_{qp} \neq T_b$, the quasiparticle number density represents a thermalized distribution of quasiparticles that is not in equilibrium with the bath. This situation can occur when energy is coupled into the electron population, and the electron–electron interaction is much stronger than the electron–phonon interaction. The changes in the quasiparticle density impact the resistivity and the kinetic inductance of the superconductor, which affects the resonator through the CPW loss and frequency shift. The quasiparticle contribution to the rf loss is⁴¹

$$\frac{1}{Q_{QP}(\omega_r, T_{qp})} = -\alpha_{Lki} \gamma \frac{\sigma_1(\omega_r, T_{qp})}{\sigma_2(\omega_r, 0)}. \quad (9)$$

Similarly, the fractional resonance frequency shift is given by

$$\frac{\delta f(T_{qp})}{f_r} = -\frac{\alpha_{Lki}}{2} \gamma \frac{\delta \sigma_2(\omega_r, T_{qp})}{\sigma_2(\omega_r, 0)}. \quad (10)$$

Here, α_{Lki} is the kinetic inductance fraction that is frequency, material, and geometry dependent, and $\omega_r = 2\pi f_r$ is the resonator frequency. The characteristic superconductor length scales are captured in the usual manner through γ that takes the values of either -1 , $-1/2$, or $-1/3$ depending on the superconductor film condition.⁴¹ The 100-nm-thick TiN film studied here satisfies the thin film condition, so $\gamma_{TiN} = -1$, while the 100-nm-thick Al film falls in the extreme anomalous limit, so $\gamma_{Al} = -1/3$. The frequency-dependent complex conductivity, $\sigma(\omega_r) = \sigma_1(\omega_r) - i\sigma_2(\omega_r)$, depends on the quasiparticle distribution function.⁴² The specific quasiparticle effective temperature is solved numerically using Eq. (7) to connect the quasiparticle rate equations to the real and imaginary components of the complex conductivity for $\omega < 2\Delta/\hbar$.⁴¹ The complex conductivity relationships are

$$\frac{\sigma_1(\omega, T^*)}{\sigma_n} = \frac{4\Delta(T^*)}{\hbar\omega} \exp\left(-\frac{\Delta(T^*)}{k_B T^*}\right) \sinh\left(\frac{\hbar\omega}{2k_B T^*}\right) K_0\left(\frac{\hbar\omega}{2k_B T^*}\right), \quad (11)$$

$$\frac{\sigma_n(\omega, T^*)}{\sigma_n} = \frac{\pi\Delta(T^*)}{\hbar\omega} \left[1 - \sqrt{\frac{2\pi k_B T^*}{\Delta(T^*)}} \exp\left(-\frac{\Delta(T^*)}{k_B T^*}\right) - 2 \exp\left(-\frac{2\Delta(T^*) + \hbar\omega}{2k_B T^*}\right) I_0\left(\frac{\hbar\omega}{2k_B T^*}\right) \right], \quad (12)$$

where σ_n is the normal state conductivity before the superconducting transition, I_0 is a zeroth-order modified Bessel function of the first kind, and K_0 is the zeroth-order modified Bessel function of the second kind. Additional details about the various material parameters are in the [supplementary material](#).

The application of this model to non-BCS superconductors would require modifications to the underlying equations that presently describe the exponential decrease of quasiparticles at low temperatures. For example, d-wave superconductors exhibit gap nodes that lead to a finite quasiparticle density even at low temperatures. As a result, the temperature dependence of key parameters such as kinetic inductance and quasiparticle number density will deviate from the behavior predicted by our model.⁴³

C. Complete model

The complete comprehensive temperature and power dependent model can be defined by combining all the different loss mechanisms,

$$\frac{1}{Q_i(\omega_r, T_b, n)} = \frac{1}{Q_A} + \frac{1}{Q_{\text{TLS}}^0} \left(\frac{\tanh \frac{\hbar\omega_r}{2k_B T_b}}{\sqrt{1 + \left(\frac{n^2}{DT_b^2}\right) \tanh \frac{\hbar\omega_r}{2k_B T_b}}} \right) - \gamma \alpha_{\text{Lki}} \frac{\sigma_1(\omega_r, T_{\text{qp}})}{\sigma_2(\omega_r, 0)}. \quad (13)$$

All the temperature and power independent loss mechanisms are grouped in the constant first term, $1/Q_A$. The second term describes the TLS loss, and the third describes the quasiparticle loss. Similarly, the fractional frequency shift can be defined by combining TLS and quasiparticle contributions,

$$\frac{\delta f(\omega_r, T_b)}{f_r} = \frac{1}{\pi Q_{\text{TLS}}^0} \left[\text{Re} \left(\Psi \left(\frac{1}{2} + \frac{1}{2\pi i} \frac{\hbar\omega_r}{k_B T_b} \right) \right) - \log \frac{\hbar\omega_r}{2\pi k_B T_b} \right] - \gamma \frac{\alpha_{\text{Lki}}}{2} \frac{\delta \sigma_2(\omega_r, T_{\text{qp}})}{\sigma_2(\omega_r, 0)}. \quad (14)$$

It is important to note that the resonance frequency, f_r , may be different than the measured resonance frequency at low temperatures if there are non-equilibrium quasiparticles present that contribute a measurable shift through the kinetic inductance.

The temperature dependent internal quality factor for a 16- μm wide centerline (8- μm wide gap) TiN resonator is presented in [Fig. 1\(a\)](#) with $\langle n \rangle = 10^4$, $Q_c = 145\text{k}$, and single-photon $Q_i = 400\text{k}$ at base temperature. Similarly, the temperature dependent internal quality factor for a 6- μm wide centerline (3- μm wide gap) Al resonator is shown in [Fig. 1\(b\)](#) with $\langle n \rangle = 10^4$, $Q_c = 151\text{k}$, and single-photon $Q_i = 412\text{k}$ at base temperature. These two plots also show data (filled circles) and model results for the total Q_i (blue solid line) as well as the calculated contributions from TLSs (red solid line) and quasiparticles (gold solid line). The right axis is used for the effective quasiparticle (black dashed line) and bath (black

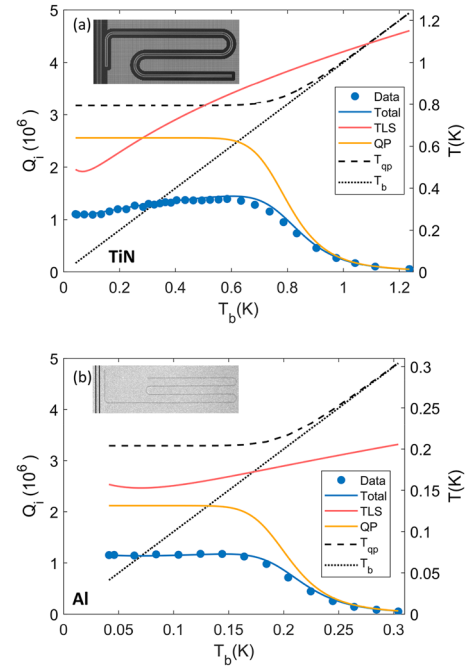


FIG. 1. Comparison of temperature dependent loss for a TiN resonator, R1 (a), and an Al resonator, R1 (b), showing data (symbols) and model (blue) results along with contributions from two-level systems (red) and quasiparticles (gold). The measured bath temperature (black dotted line) and the calculated quasiparticle temperatures (black dashed line) are also shown. The inset shows the optical image of the resonators. The TiN resonator has a 16- μm wide centerline with an 8- μm wide gap, while the Al resonator has a 6- μm wide centerline with a 3- μm wide gap.

dotted line) temperatures from the model that describes these data. At base temperature, the power dependence of these resonators shows no saturation even at higher photon numbers, indicating that the power-independent loss does not contribute to quality factor measurements reported here. Therefore, Q_A is removed from further consideration.

According to these model results, at high bath temperatures, the quasiparticles are thermalized and $T_{\text{qp}} = T_b$. At lower temperatures, the quasiparticles are not in equilibrium with the phonons, resulting in $T_{\text{qp}} > T_b$. For TiN, the quasiparticle's effective temperature, $T_{\text{qp}} \sim 794$ mK, corresponds to an $n_{\text{qp}} = 282 \mu\text{m}^{-3}$. When compared to the BCS Cooper pair number density,³³ $n_{\text{cp}} = N(0)\Delta(0)$, the relative carrier density ratio is $n_{\text{qp}}/n_{\text{cp}} \sim 1.19 \times 10^{-5}$. For Al, $T_{\text{qp}} = 204$ mK, which corresponds to an $n_{\text{qp}} = 103 \mu\text{m}^{-3}$ and a carrier density ratio $n_{\text{qp}}/n_{\text{cp}} = 3.1 \times 10^{-5}$. It is interesting that the quasiparticle number density is comparable for these two resonators, suggesting that the excess energy is coupled to both samples in a repeatable manner, similar to that from an imperfect experimental setup. The greater impact of the quasiparticle loss on the Al resonator is a result of the smaller superconducting gap compared to TiN and is evident through the calculated contributions and relative Q associated with the TLSs and quasiparticles.

III. RESULTS AND DISCUSSION

The superconducting resonators are measured in transmission mode on the low-temperature stage in an adiabatic diamagnetic refrigerator, configured with 73.2 dB of total attenuation at various thermal plates on the input line. DC blocks are installed at 1 K on both the input and output lines. On the output line, a three-stage rf isolator and low-noise amplifier are installed on the 3 K plate. The transmission spectrum for each resonator is collected by stepping through temperatures and, at each temperature, measuring S_{21} for a range of different applied powers, resulting in the average photon numbers, $\langle n \rangle$, varying from those corresponding to approximately one photon in the resonator to powers just before the onset of nonlinear effects.^{44,45} The internal quality factor is extracted from the measured complex transmission spectra. The power dependent internal quality factor is interpolated for each constant-temperature series to provide data at uniform average photon numbers. The temperature dependence of the quality factors and frequency shifts is calculated using Eqs. (10) and (11). Details related to data collection and interpolation are provided in the [supplementary material](#). The temperature and power dependent internal quality factor and fractional frequency shift data and model of two TiN resonators are shown in Fig. 2.

The internal quality factor in Figs. 2(a) and 2(c) has three primary temperature regions. In the high-temperature region, losses are dominated by thermally generated equilibrium quasiparticles. The elevated temperatures result in an increased density of thermal quasiparticles where $T_{qp} = T_b$, resulting in a decreasing Q_i . For intermediate temperatures, the transition from thermal quasiparticle loss into loss dominated by TLS and non-equilibrium quasiparticles is characterized by the “knee” near 700 mK and the highest values of

Q_i . The low-temperature regime shows varying behavior depending on the average photon number. Above 100 mK, the typical TLS thermal behavior dominated by $\tanh(\hbar\omega/2k_B T)$ in Eq. (1) is observed. For low powers, there is an increased lift in Q_i at the lowest temperatures, but at high powers, there is a drop, which is reflected as a change in sign of β_1 that indicates a change in TLS–TLS interactions. These trends have been observed in other materials.^{24,46} Owing to their empirical nature, D and β_1 are chosen to be power dependent parameters, which provide the necessary degree of freedom to achieve the fits shown in Figs. 2(a) and 2(c). Their variation from resonator to resonator depends on the specific arrangement of TLS defects.

The fractional frequency shifts shown in Figs. 2(b) and 2(d) are power independent, and the three regimes observed for the internal quality factor are also observed. This behavior is captured by the model. It should be noted that the number of fit parameters needed to fit the fractional frequency shift is fewer than those needed to fit the internal quality factor. Therefore, the fractional frequency shift is fit first, and those parameters are then used to fit the internal quality factor as described in the [supplementary material](#).

For the aluminum resonator results shown in Fig. 3, the relative contributions of the quasiparticles and TLS defects in the low-temperature region are evident. The data are flatter in both internal quality factor and fractional frequency shift compared to the TiN resonator that has a larger T_c . This flatness is caused by the significant quasiparticle contributions that are either comparable to or larger than the TLS contributions at different photon powers. It is important to note that these observations are most likely limited by the thermalization of the experimental setup and may be impacted by incomplete shielding or the short delay between cooling the sample and starting measurements in an adiabatic

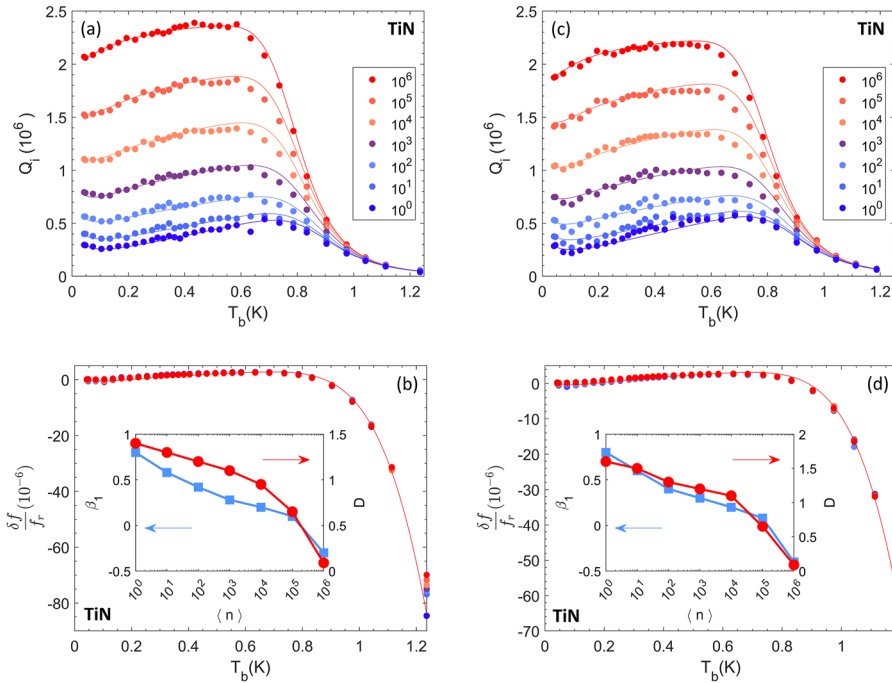


FIG. 2. For TiN resonators, temperature and power dependent internal quality factor [(a) and (c)] for different n indicated in the legend and fractional frequency shift [(b) and (d)] data (symbols) and model (lines) of two resonators, R1 [(a) and (b)] and R2 [(c) and (d)]. The inset graph shows power dependent fitting parameters associated with TLS defects for β_1 (blue squares) and D (red circles); lines are used to guide the eye.

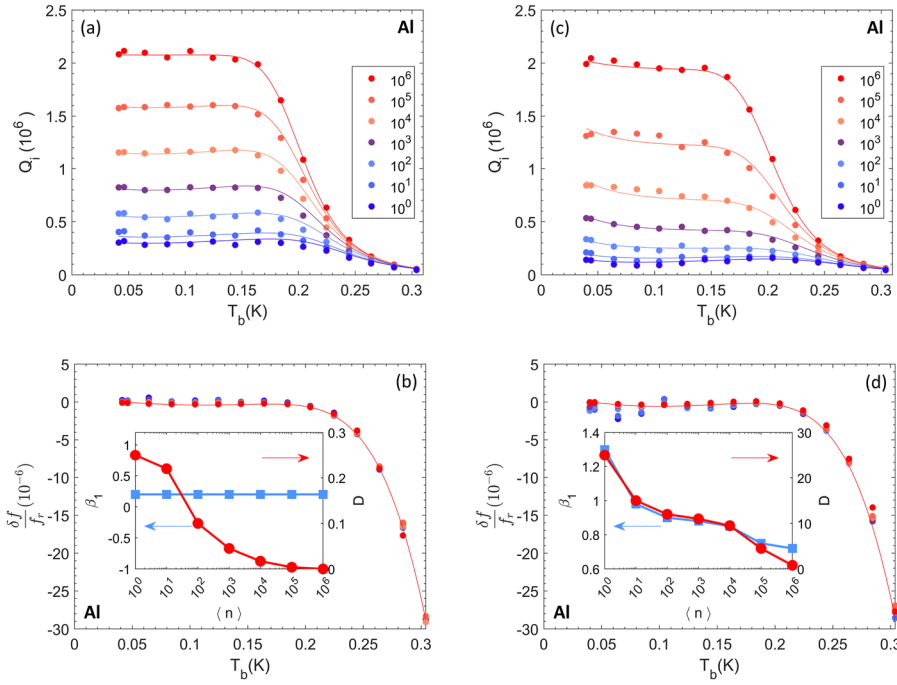


FIG. 3. For Al resonators, temperature and power dependent internal quality factor [(a) and (c)] for n indicated in the legend and fractional frequency shift [(b) and (d)] data (symbols) and model (lines) of two resonators, R1 [(a) and (b)] and R2 [(c) and (d)]. The inset graph shows power dependent fitting parameters associated with TLS defects for β_1 (blue squares) and D (red circles); lines are used to guide the eye.

demagnetization refrigerator that may lead to incomplete cooling of the sample and rf components.

It is useful to acknowledge that by itself, the flat, low-temperature behavior of the R1, Q_i , is insufficient to isolate the potential limit imposed by constant loss, Q_A , from the nonequilibrium QP loss identified here. Qualitative comparisons of the subtle rounding of the transition (“knee” or “elbow”) at mid-temperatures that softens in the presence of quasiparticles is insufficient to isolate the potential impact of Q_A . Instead, clarity is provided with this model through the fractional frequency shift that has a flat frequency behavior at low temperatures from quasiparticles and complex frequency dependence from the TLS defects. Other mechanisms that might be considered, such as external flux and radiative/packaging loss, would need to modify the complex conductivity to impact both loss and frequency observables in a consistent manner to lower confidence in the assessed model values.

Similar increases in the quality factor at low temperatures that were observed for the TiN resonator have been reported for aluminum resonators.⁴⁶ The power dependence of the aluminum resonators in the low and mid temperature regimes is attributed to the power dependence of TLS defects. The comparable contributions of the two sources of loss can cause the power dependent loss assessments to have a lower measure Q and a weaker power dependence.

All fit parameters for these resonators are reported in Table I. Given the complexity of the model and number of non-orthogonal parameters, uncertainty analysis has not been performed on the model results. As is evident in both resonators, the model has the most difficulty representing the trends in the intermediate temperature region. It is essential to point out that the extracted values of I_{ext} are not absolute but depend on the choice of T_s . For the four

resonators presented here, the lowest calculated temperatures are largely power independent. The critical temperature and resonator frequency are measured parameters, while the rest are fit parameters. Of the parameters that are fit to the model, only the kinetic inductance fraction is directly observable. To qualify the low values of the kinetic inductance, the rf kinetic inductance is estimated from the geometric inductance of coplanar waveguides that do not have trenches. The calculated values for TiN are 0.93 pH/ $\square$ and 0.31 pH/ $\square$ for Al. These values are relatively comparable to the kinetic inductance determined from dc transport measurements and BCS theory to be 1.9 pH/ $\square$ for TiN and 0.085 pH/ $\square$ for Al. Additional details are in the [supplementary material](#).

The empirical description of TLS defect loss adopts the form from Crowley *et al.*²⁴ Of interest is the relationship between the internal quality factor and potential trends in the relative TLS decoherence time, $T_{2\text{TLS}} \propto (DT^{\beta_1})^{-1}$. The temperature dependent trends of the TLS decoherence time are shown in Fig. 4.

There are three key observations from these two graphs. The first is a comparison between resonator materials and the relative contribution of quasiparticles that tend to flatten out the temperature dependence. This is observed for high temperatures for the resonators in Figs. 4(a), 4(b), and 4(d), and all temperatures and powers for the Al resonator R1 in Fig. 4(c). The key is the different T_C of these two materials and the relatively high I_{ext} from the experimental setup. The general trend of decreasing decoherence time for increasing temperature is consistent with previous observations.⁴⁷

The second observation is that the TLS decoherence time first decreases and then increases with power at lower temperatures for TiN. This behavior might be the result of the TLS population being in the ground state with a smaller interaction strength at low powers, resulting in a relatively high TLS decoherence time that is

TABLE I. Comparison of TLS loss parameters determined by a power dependent TLS-only model and a power-temperature dependent TLS-quasiparticle model for this TiN resonator. Parameters marked with an * are determined by experiments.

Parameter	TiN R1	TiN R2	Al R1	Al R2
Q_{TLS0}	0.11×10^6	0.11×10^6	0.12×10^6	0.08×10^6
D	{1.4, 1.3, 1.2, 1.1, 0.95, 0.65, 0.09}	{1.6, 1.5, 1.3, 1.2, 1.1, 0.65, 0.09}	{0.25, 0.22, 0.1, 0.045, 0.017, 0.0038, 2×10^{-5} }	{25, 15, 12, 11, 9.5, 4.5, 0.80}
β_1	{0.8, 0.58, 0.42, 0.28, 0.2, 0.1, -0.3}	{0.8, 0.6, 0.4, 0.30, 0.20, 0.08, -0.4}	0.2	{1.3, 0.98, 0.90, 0.88, 0.85, 0.75, 0.72}
β_2	0.55	0.55	0.15	0.57
$I_{ext} (\mu m^{-3} s^{-1})$	2.37×10^8	2.75×10^8	3.26×10^7	2.97×10^7
Γ_S (MHz)	0.82	0.82	0.31	0.31
α_{Lki}	0.123	0.123	0.108	0.108
f_r (GHz)*	5.0186	4.4695	5.2508	4.7371
T_C (K)*	5.29	5.29	1.25	1.25

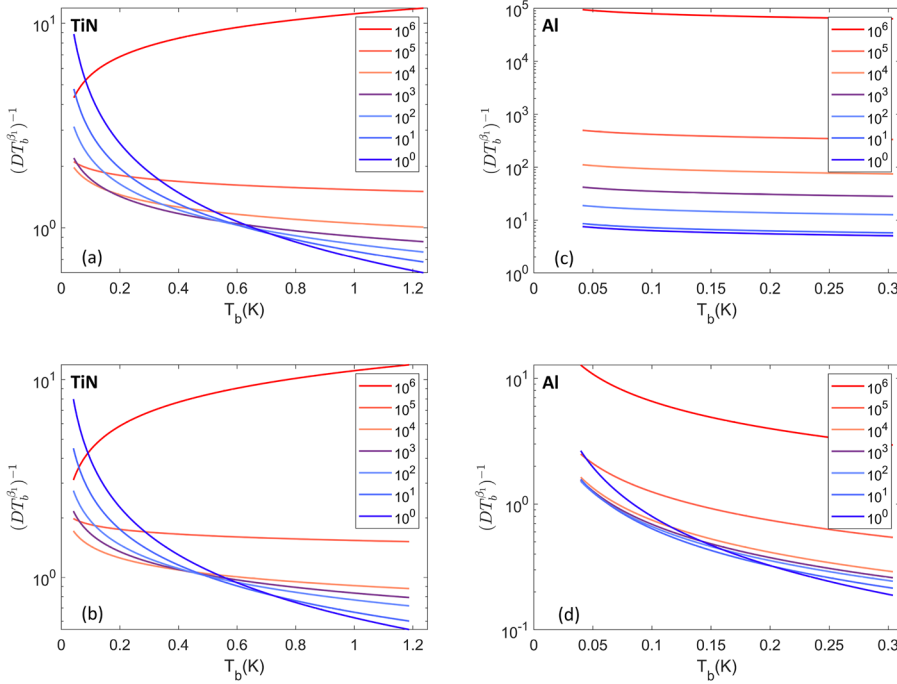


FIG. 4. Comparison of the temperature and power dependent TLS decoherence time ($T_{2TLS} \propto (DT_b^{\beta_1})^{-1}$) for the TiN resonators R1 (a) and R2 (b) and the Al resonators R1 (c) and R2 (d) for different average photon numbers as shown in the legend.

complemented by a region of high decoherence time when the TLS population is becoming polarized. It is also possible that athermal quasiparticles contribute to these trends.

In addition, this model implements independent contributions of TLS and quasi-particles to the loss and frequency shift of resonators. While it is possible that they may be coupled through the phonon bath, a potential common mode of relaxation, the observables explored here are insufficient to provide the required insight. Instead, spectroscopic and time-dependent studies may be more appropriate.

The third observation is the change in TLS decoherence time at the highest temperatures, where the TLS decoherence time increases with power. In this region, it is possible that the TLS bath is becoming saturated and polarized as applied power is increased.

These effects may also result from athermal quasiparticles, where increases in Q_i have been observed in quasiparticle-limited systems.^{20,48} To better understand this behavior, a completely different model that describes the microscopic behavior of non-thermal quasiparticle dynamics is needed.

The results and analysis presented here are indeed complex. There are over a hundred measurements reported on each resonator under different temperatures and powers. There are many free parameters, so a structured fitting process is used to work through the parameters to eliminate arbitrary numerical results. First, using alternative experiments to determine baseline observable parameters such as T_C and f_r , as well as literature values for fundamental materials parameters such as Γ_R , are used to reduce the number of parameters. Next, the fractional frequency

shift is fit because it has fewer parameters than the internal quality factor, enabling the determination of $\alpha_{L_{ki}}$ and Q_{TLS}^0 . Only then is the quality factor fit to determine the remaining parameters, using the high-power results to refine Γ_S . For the results presented here, there is no constant loss considered, $Q_A \gg Q_{QP}, Q_{TLS}$. A key feature of this model and process is that there is consistency of the quasiparticle parameters across resonators on a single chip, as expected from a homogeneous material.

It is common for superconducting resonators to be measured under a smaller range of conditions, such as low-temperature power-sweeps or high-power temperature-sweeps, to maximize measurement efficiency. The results of such measurements are often analyzed with parts of the model presented here. The successful application of the model on all the data collected from the Al and TiN resonators with a single set of parameters across loss and frequency shift demonstrates the utility of the complete model, allowing aspects of the TLS–TLS behavior to be more holistically assessed.

IV. CONCLUSION

In conclusion, understanding, disentangling, and mitigating the losses from two-level systems and quasiparticles is an important topic for quantum information science. The power and temperature dependent microwave loss and frequency shift model presented describes the loss and frequency shift from two resonators made from different materials over a wide range of experimental conditions. The small set of parameters needed to describe loss and frequency shift across all measured conditions has value to extract parameters that represent the sample, helps reduce misinterpreting two-level system and quasiparticle contributions, and may help characterize TLS–TLS interactions. These results may help data analysis on various experimental configurations and different BCS superconducting materials and assist in determining strategic needs for improving either device fabrication or measurement.

SUPPLEMENTARY MATERIAL

The [supplementary material](#) includes detailed information on the resonator fabrication and measurement setup, Q-factor extraction, data collection, and a discussion of the material parameters used in the temperature- and power-dependent model.

AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Ashish Alexander: Conceptualization (equal); Data curation (lead); Formal analysis (lead); Investigation (lead); Methodology (equal); Validation (equal); Visualization (lead); Writing – original draft (lead); Writing – review & editing (equal). **Christopher G. Weddle:** Investigation (supporting). **Christopher J. K. Richardson:** Conceptualization (equal); Data curation (supporting); Formal

analysis (supporting); Methodology (equal); Supervision (lead); Validation (equal); Visualization (supporting); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon the reasonable request.

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